

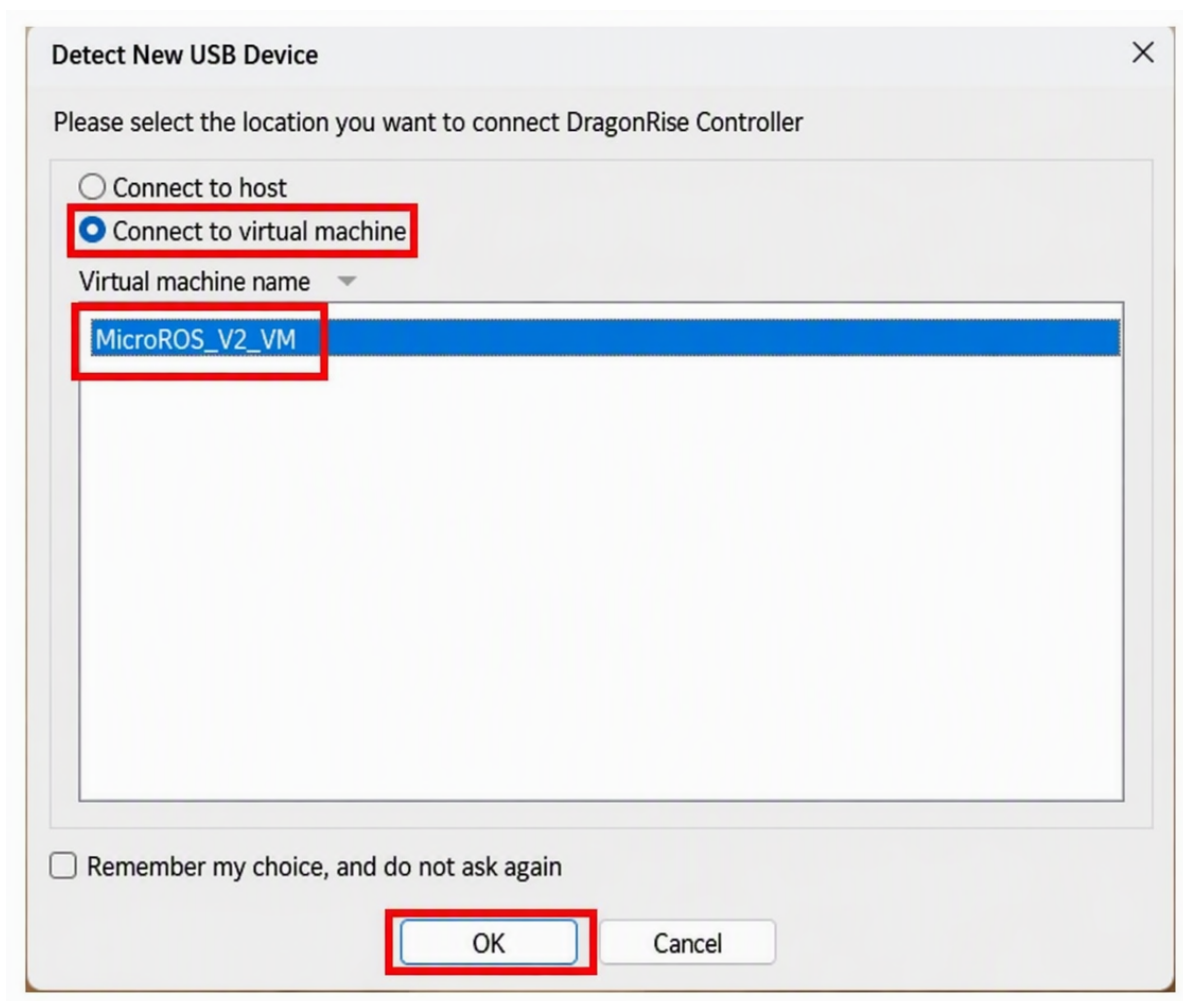
Quick-Start Handle Control in the Virtual Machine

1. Course Content

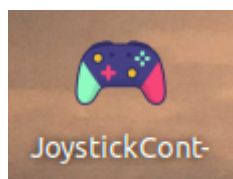
Use a USB gamepad to control the robot car's movement and buzzer, providing a more intuitive operating experience than the keyboard.

2. Preparation

- Plug the USB gamepad receiver into the computer and choose to connect the USB device to the virtual machine



- Double-click the **JoystickControl** icon on the virtual machine desktop to quickly start gamepad control



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JoystickControl
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[INFO] [launch]: All log files can be found below /home/yahboom/.ros/log/2026-07-27-10-26-14-330411-yahboom-144690
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [micro_ros_agent-1]: process started with pid [144744]
[INFO] [base_driver-2]: process started with pid [144747]
[INFO] [joy_node-3]: process started with pid [144750]
[INFO] [joy_control_node-4]: process started with pid [144752]
[micro_ros_agent-1] [1785119175.252180] info      | UDPv4AgentLinux.cpp | init      | running...
| port: 8090
[base_driver-2] [INFO] [1785119175.519403841] [base_driver]: === BaseDriverNode Parameters ===
[base_driver-2] [INFO] [1785119175.519703924] [base_driver]:   odom_linear_speed      : 0.15 m/s
[base_driver-2] [INFO] [1785119175.519942031] [base_driver]:   odom_angular_speed    : 0.5 rad/s
[base_driver-2] [INFO] [1785119175.520106709] [base_driver]:   odom_position_tolerance : 0.02 m
[base_driver-2] [INFO] [1785119175.520292035] [base_driver]:   odom_angle_tolerance   : 0.05 rad
[base_driver-2] [INFO] [1785119175.520460301] [base_driver]:   odom_timeout           : 30.0 s
[base_driver-2] [INFO] [1785119175.520637336] [base_driver]:   min_linear_speed       : 0.03 m/s
[base_driver-2] [INFO] [1785119175.520793649] [base_driver]:   min_angular_speed      : 0.2 rad/s
[base_driver-2] [INFO] [1785119175.520949146] [base_driver]:   precise_adjust_xy_pid  : [0.5, 0.0, 0.05]
[base_driver-2] [INFO] [1785119175.521153669] [base_driver]:   precise_adjust_theta_pid: [1.0, 0.0, 0.3]
[base_driver-2] [INFO] [1785119175.521326778] [base_driver]:   precise_adjust_max_lin  : 0.12 m/s
[base_driver-2] [INFO] [1785119175.521487652] [base_driver]:   precise_adjust_max_ang  : 0.5 rad/s
[base_driver-2] [INFO] [1785119175.521648957] [base_driver]:   precise_adjust_pos_tol  : 0.01 m
[base_driver-2] [INFO] [1785119175.521796698] [base_driver]:   precise_adjust_ang_tol  : 0.06 rad
[base_driver-2] [INFO] [1785119175.521937616] [base_driver]: =====
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3. Gamepad Key Bindings

SELECT (Only valid for ROS2 version of the handle control program)

Single press: Enable/disable handle ROS2 program control

Arrow keys

Up/Down: Forward/Backward

Left/Right: Left/Right

START: Press the
buzzer to sound,
Release the buzzer
no sound

(For robot car with PTZ)

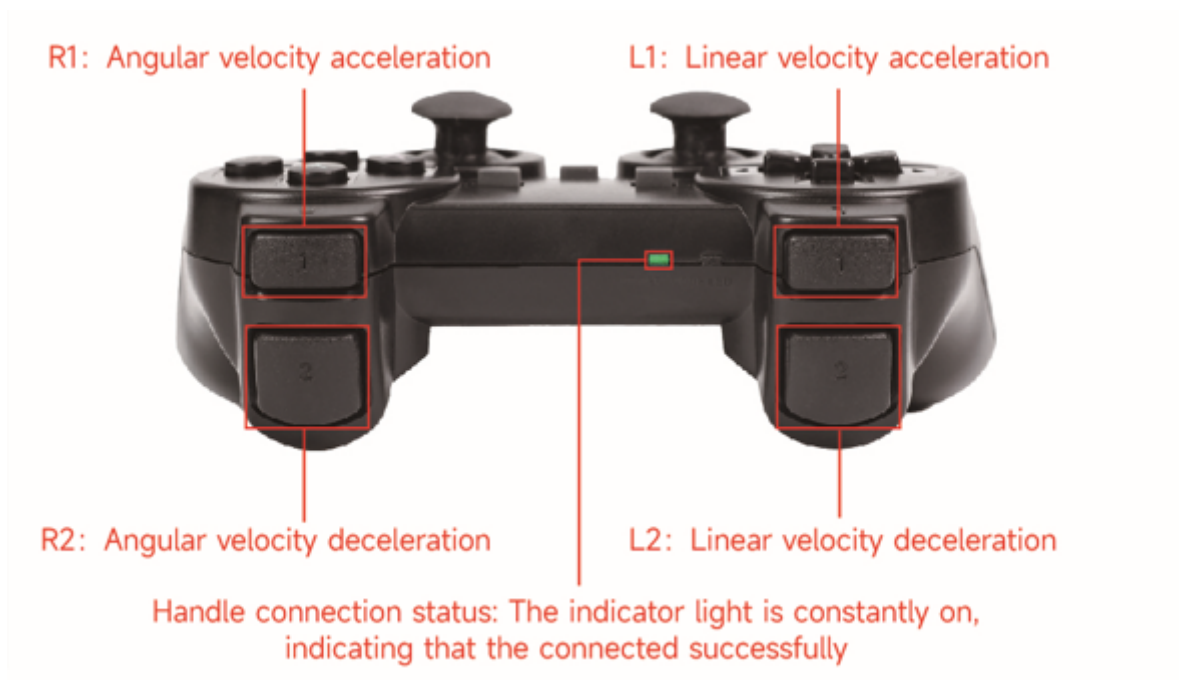
Y: PTZ upwards

A: PTZ downwards

X: PTZ turn left

B: PTZ turn right





3.1 Movement Control

Control uses per-axis arbitration between the joysticks and the D-pad — the joystick takes priority, and the D-pad acts as a fallback only when the corresponding joystick axis is idle:

Input	Function	Curve	Description
Left stick up/down (LY)	Forward / Backward	Square curve (power=2.0)	Analog control with a 0.15 dead zone
Right stick left/right (RX)	Turn left / Turn right	Cubic curve (power=3.0)	Analog control with a 0.15 dead zone
Left stick left/right (LX)	No function	—	Unused
Right stick up/down (RY)	No function	—	Unused
D-pad up/down	Forward / Backward	—	Takes effect only when LY is idle (digital fallback)
D-pad left/right	Turn left / Turn right	—	Takes effect only when RX is idle (digital fallback)

3.2 Speed Levels

There are 5 fixed speed levels (G1–G5), with G2 as the default:

Level	Linear speed	Angular speed
G1	0.20 m/s	0.50 rad/s
G2 ★	0.30 m/s	1.50 rad/s
G3	0.50 m/s	2.50 rad/s
G4	0.75 m/s	3.75 rad/s
G5	1.00 m/s	5.00 rad/s

Key	Function
L1	Increase linear speed by one level
L2	Decrease linear speed by one level
R1	Increase angular speed by one level
R2	Decrease angular speed by one level

Reaching G5 or dropping to G1 triggers a double beep.

3.3 Handle Control

Key	Function	Description
X	Servo S1 forward	Effective when X is pressed alone
B	Servo S1 reverse	Effective when B is pressed alone
Y	Servo S2 reverse	Effective when Y is pressed alone
A	Servo S2 forward	Effective when A is pressed alone

S1 travel range: $-90^{\circ} \sim +90^{\circ}$; S2 travel range: $-90^{\circ} \sim +20^{\circ}$.

3.4 Buzzer and Enable Switch

Key	Function
START (OPTIONS)	Hold to beep continuously; release to stop
SELECT (SHARE)	Toggle the gamepad enable state (/Joystate : True ↔ False)